



A deep neural network using modified EfficientNet for skin cancer detection in dermoscopic images

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ARTICLE INFO

Keywords:

Classification
Deep learning
EfficientNet
Fine-tuning
Skin cancer
Transfer learning

ABSTRACT

Artificial intelligence (AI) systems can assist in analyzing medical images and aiding in the early detection of diseases. AI can also ensure the quality of services by avoiding misdiagnosis caused by human errors. This study proposes a deep neural network (DNN) model with fine-tuned training and improved learning performance on dermoscopic images for skin cancer detection. A knowledge base for training the DL models is constructed by combining different dermoscopic datasets. Transfer learning and fine-tuning are implemented for faster training of the proposed model on a limited training dataset. The data augmentation techniques are applied to enhance the performance of the model. A total of 58,032 refined dermoscopic images were used in this study. The output of the layered architecture is aggregated to perform the binary classification for skin cancer. The performance of the trained models is investigated for multiclass and binary classification tasks. The performance metrics confirm that the DNN network with modified EfficientNetV2-M outperforms the state-of-the-art deep learning-based multiclass classification models.

1. Introduction

Excessive ultraviolet (UV) exposure on the skin causes mutation and malfunction of melanocytes, resulting in skin cancer [1–3]. Another major factor oriented for the growth of skin cancer cells is based on the patient's skin color and age [4,5]. The severity of skin cancer can be fatal when the condition is left unobserved. The abnormal occurrence of any lesions on the skin needs to be diagnosed as skin cancer. The skin lesions can be classified as malignant, indicating skin cancer, and benign, representing non-cancerous. In the conventional screening process, the dermatologist analyzes the skin lesion with the help of a dermatoscope [6–8]. Based on the dermatologist's skill, the skin lesion is classified. Furthermore, if the dermatologist diagnoses the lesion as malignant, the patient is advised to undergo a biopsy test for confirmation. A biopsy is considered the gold standard for diagnosing skin cancer because it allows for the direct examination of tissue samples under a microscope, which can confirm or rule out the presence of certain types of skin cancer. Additionally, skin lesions can be difficult to diagnose even for experienced dermatologists using dermoscopy and may be prone to misdiagnosis [9–11]. The main limitation of biopsy is that it is an invasive method where a small sample of tissue from the skin lesion is taken for examination, which can be painful and may leave a scar. The biopsy test can be costly and time-consuming. To overcome these challenges, computer-aided

diagnosis (CAD) systems are widely used to assist dermatologists in making informed decisions [12–15].

Artificial intelligence (AI) and machine learning (ML) development have paved the way for new possibilities in assistive diagnosis in the fields of biomedicine and healthcare [16,17]. In general, convolutional neural network (CNN) based systems are widely used in medical imaging for classification and segmentation applications [18–20]. From recent research studies by Esteva et al. [21] and Haenssle et al. [22], the CNN models outperformed a group of certified dermatologists for skin lesion classification task. The International Skin Imaging Collaboration (ISIC) worked to collect the dataset for dermoscopic images and conducted various challenges and events for validating the performance of newly developed algorithms and models in the diagnosis of skin cancer [23].

1.1. Related works

Many deep learning (DL) based skin cancer diagnosis methods have been proposed in the literature. Kadampur et al. [24] developed an application-based model using the Squeezenet model [25], and their model achieved an average AUC score of 0.99 on evaluation with 10% of the HAM10000 dataset [26]. The ensemble-based learning approach on multiple models proposed by Rahman et al. [27] using

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<https://doi.org/10.1016/j.dajour.2023.100278>

Received 2 December 2022; Received in revised form 2 May 2023; Accepted 21 June 2023

Available online 24 June 2023

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DenseNet, ResNet, Xception, SeResNeXt, and ResNeXt achieved the best micro average recall of 0.94 for multiclass classification when evaluated on ISIC 2019 [26,28,29] and HAM10000 [26] datasets. Chaturvedi et al. [30] developed a modified ResNeXt-101 to classify the skin lesions into seven classes. They fine-tuned their models on 8,9213 dermoscopy images and obtained an accuracy of 93.20% on the HAM10000 [26] dataset. Pacheco et al. [31] proposed an ensemble of three CNN models named PNASNet, ResNet-152, and VGG-19, along with the patient's meta-data. Their proposed method obtained a balanced accuracy of 90.10% on ISIC 2019 [26,28,29] test dataset. Similarly, Ha et al. [32] proposed ensemble-based learning using the EfficientNets [33], demonstrating the highest performance among the deep neural network (DNN) models for ISIC 2020 [34] challenge. Their model attained an AUC score of 0.96. Khan et al. [35] developed an ensemble-learning framework using the EfficientNet-B3 model for analyzing skin lesions and Extreme Gradient Boosting (XGB) for clinical data to diagnose skin cancer. Their study used the PAD-UFES-20 [36] dataset with six unbalanced categories of skin cancer and applied data augmentation [37] to overcome the imbalance. They reported that the integration of clinical data with skin lesions improves the accuracy of automated diagnosis. Their proposed model outperformed the previous studies on the PAD-UFES-20 dataset with an accuracy of 0.78, precision of 0.89, recall of 0.86, and F1 score of 0.88. Ali et al. [38] used transfer learning [39,40] and fine-tuning on pre-trained ImageNet weights [39,40] to train EfficientNet variants (B0-B7) [33] on the HAM10000 [26] dataset for multiclass skin cancer classification. They reported that EfficientNet-B4 achieved an F1 Score of 0.87 and an accuracy of 87.91%. Sukanya et al. [41] segmented lesions using the Self Adaptive Sea Lion Algorithm and used a "Deep Belief Network (DBN)" to classify the dermoscopy images into melanoma and benign. They obtained an accuracy of 94.00% on the HAM10000 [26] dataset. Zafar et al. [42] used pre-trained DesneNet-201, and the features are extracted using Slim Mould Algorithm (SMA) and is given as input to SVM and KNN classifiers. The authors reported an overall accuracy of 91.7%, 92.01%, 98.88%, and 99.33% on ISIC 2019 [26,28,29], HAM10000 [26], PH2 [43], and MED-NODE [44] datasets, respectively. Jaisakthi et al. [45] developed deep CNN models to classify skin cancer into melanoma and benign using ISIC 2019 [26,28,29] and ISIC 2020 [34] datasets. They used transfer learning [39,40], data augmentation [37], and metadata are used to improve classification accuracy. The models were optimized using the ranger optimizer, and their experiments show that the EfficientNet variants perform better than other DL models. They reported an AUC score of 0.94 using fine-tuned EfficientNet-B6 model. Recently, Gilani et al. [46] proposed a spiking VGG-13 model using the surrogate gradient descent method. The proposed model achieved an accuracy of 89.50% on ISIC 2019 [26,28,29] test dataset. They reported that their model is power efficient due to fewer trainable parameters and can be implemented in hardware.

1.2. Research gaps and motivation

The identified research gaps concerning multiclass skin lesion classification include the fact that the majority of research works focus on binary classification of lesions, i.e., melanoma or benign. Furthermore, most of the multiclass classification models are trained on limited datasets. Thus, to enhance the performance of DL-based skin lesion classification models, they need to be trained and analyzed using a larger dataset. In addition, limited research works are available in the literature that employs transfer learning and fine-tuning strategies on pre-trained DL models with layered architecture for the classification of dermoscopic images.

From the literature, it is evident that EfficientNet models have been used widely by researchers recently. Also, EfficientNet-based classification models have displayed better results than other state-of-the-art CNN models. This motivated us to use the layered architecture of EfficientNet-B4 and the latest EfficientNetV2-M models. The EfficientNetV2 is a new family of EfficientNets with faster training speed and better parameter efficiency than their previous versions [47].

1.3. Research contributions

The major contributions of this study are outlined below:

- We have created modified versions of the EfficientNet V2-M and EfficientNet-B4 baseline models by altering the final fully-connected layer and the softmax layer, enabling the classification of skin lesions into seven distinct classes.
- The developed models have been trained using a comprehensive knowledge base generated by amalgamating dermoscopy images from the ISIC 2020 [34], ISIC 2019 [26,28,29], and ISIC 2018 [48] datasets, following the data-centric strategies proposed by Cassidy et al. [49]. The resulting knowledge base comprises seven targeted classes.
- Various efficient training strategies, including fine-tuned transfer learning [39,40] and various image augmentation [37], were employed in this study.
- The proposed model can accurately identify the significant features in dermoscopic images of skin lesions and classify them into seven major classes. The models' performance for multiclass and binary classification has been evaluated using accuracy, precision, recall, and F1 score.
- We have compared the performance of the modified EfficientNet-V2-M model with the latest state-of-the-art DL models employed for multiclass skin lesion classification.

1.4. Content of the paper

The rest of the paper is structured into four sections. Section 2 elaborates on constructing the proposed method with DNN layered architecture leveraging the transfer learning technique. The training of the proposed model with a comprehensive knowledge base is discussed. Section 3 evaluates the model in a data-centric approach on HAM1000 [26], ISIC 2019 [26,28,29], and ISIC 2020 [34] test datasets. The performance metrics are captured and discussed using standard evaluation metrics. The performance of the proposed model is compared with 14 state-of-the-art DL models, and the model behavior is explained using Grad-CAM. Finally, Section 4 summarizes the observations and conclusions with future directions.

2. Methodology

The proposed method consists of a DNN system trained on a comprehensive knowledge base. The high-level blocks are interconnected, as shown in Fig. 1. The DNN system is trained with information from the knowledge base labeled with seven classes. The output of the DNN system is aggregated into binary classes, namely malignant and benign lesions. These binary classes are the parent class for the labeled skin lesions in the dataset. The implemented DNN system includes a DNN model with transfer learning [39,40] for feature extraction from the dermoscopic images and a classifier layer to predict the class labels.

2.1. Knowledge base

The proposed method is fine-tuned and trained with a large combined dataset from ISIC 2020 [34], ISIC 2019 [26,28,29], and ISIC 2018 [48] dermoscopic images. The data with unique information across the dataset is captured during the data cleaning process. The cleaning process filters the duplicated and invalid images across the dataset and categorizes the data with the skin lesion. It is observed that the dermoscopic images are labeled into seven major classes that are actinic keratosis (AK), basal cell carcinoma (BCC), benign keratosis-like lesions (BKL), melanoma (MEL), melanocytic nevus (NV), squamous cell carcinoma (SCC), and unknown [27]. The labels are provided by the medical experts based on the nature of skin lesions. These seven classes can be broadly categorized into malignant and benign lesions. The MEL, BCC, and SCC are grouped as malignant class, and the AK,

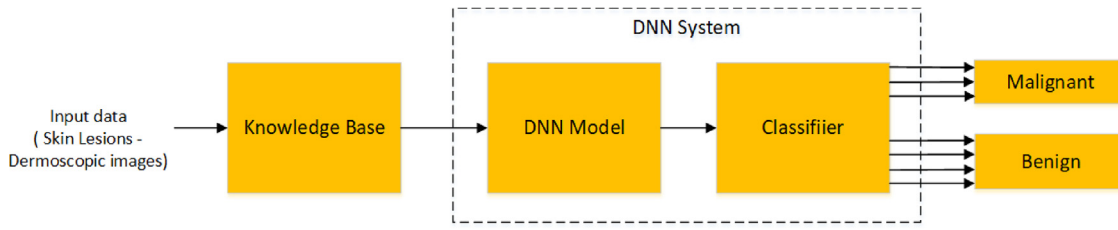


Fig. 1. High-level block diagram of the proposed model.

Table 1

Image statistics across the different datasets used in creating the knowledge base.

Database name	Malignant			Total	Benign				Total
	MEL	BCC	SCC		AK	BKL	NV	Unknown	
ISIC 2020	581	0	0	581 (1.78%)	0	221	5191	26 708	32 120 (98.22%)
ISIC 2019	4522	3323	628	8473 (33.45%)	867	2624	12 875	492	16 858 (66.55%)
ISIC 2018	1113	514	0	1627 (16.24%)	327	1099	6705	257	8388 (83.76%)
HAM10000	1113	514	0	1627 (16.24%)	327	1099	6705	257	8388 (83.76%)
Combined dataset	5103	3323	628	9054 (15.60%)	867	2845	18 066	27 200	48 978 (84.40%)

BKL, NV, and unknown type of lesions are classified as benign. Thus, binary classes of lesions, namely malignant and benign, are created in the combined dataset. The combined dataset consists of 58,032 images captured using dermoscopy. These images are preprocessed and augmented by image blurring, distortion, resizing, and normalization techniques to improve the data distribution [37]. The statistics of different datasets used to construct the combined dataset, which is used as the knowledge base for training the models, are shown in Table 1. A significant class imbalance is observed across the datasets, which may affect the model generalization capability. However, the combined dataset yields a stronger knowledge base to improve the performance of the DL models. Also, the progressive learning in the EfficientNet models improves the regularization of the model and prevents overfitting [47].

2.2. DNN system

The proposed deep layered model is implemented using a transfer learning [39,40] mechanism for detecting skin cancer in dermoscopic images. The transfer learning technique on CNN improved the training performance by extracting the image feature efficiently when compared to training the models from scratch [39,50,51]. The proposed DNN models are created by modifying EfficientNet-B4 and EfficientV2-M baseline models. The layered architecture in the EfficientNets family is considered to be optimal based on the trade-off between the learning parameter involved in the architecture and the achievable accuracy on the ImageNet dataset. This DNN architecture is used to build the fine-tuned feature extraction system in the proposed model. The extracted features are propagated into a neural classifier model consisting of two dense layers of 500 and 256 neurons, respectively, followed by output layers for seven targeted classes. In subsequent stages, these target classes are aggregated into binary classes, namely malignant and benign, to provide high-level skin cancer detection. The entire process in the proposed method with the DNN layered model architecture is depicted in Fig. 2.

The learning weights and biases from the model trained with the ImageNet dataset are initialized for effective image feature extraction in transfer learning [39]. The proposed model is fine-tuned and trained for skin lesions with seven target classes (AK, BCC, BKL, MEL, NV, SCC, and unknown). Training the individual skin lesion helps the model to understand the hidden features of the dermoscopic images. The hyperparameters used in the proposed model are shown in Table 2. The suitable learning parameters are considered for adjusting the learning weights and bias during transfer learning. The learning rate determines the attention of the model to detailed information about the features of lesions belonging to different classes. The training iteration may

Table 2

Hyperparameters configuration used in the proposed DNN network.

Hyperparameters	Values
Image size	224 × 224
No. of epoch	16
Learning rate	3×10^{-5}
Learning rate scheduler	Cosine decay [52]
Learning warm up epoch	1
Training batch size	4
Optimizer	Adam [53]
Training validation split	80:20

Table 3

System specifications used to train the proposed model.

System	Configuration
Hardware	Host CPU architecture
	Host CPU RAM
	External GPU
Software	AI framework
	CUDA toolkit
	DNN runtime library
	Development IDE
	Intel core i5 - 9th gen
	16 GB
	NVIDIA GTX 1660Ti
	Tensorflow v2.6.0
	11.4
	cuDNN v7.6.5
	Jupyter Lab

increase for converging to the global loss minima for a very small learning rate. Whereas, for a higher learning rate, the training may diverge the ideal global minima and tend the model to display a significantly larger loss value. The learning rate scheduler provides a predetermined framework approach for tuning the learning rate during training iterations. The cosine decay annealing [52] is adopted as the learning rate scheduler in the proposed work. A larger learning rate is given in the initial epochs, termed warm-up epoch, and decreases before it increases again to achieve the minimum loss in a short number of epochs [52]. This technique helps the model to learn discriminant features and prevents the model from being overfitted on the training dataset. The Adam optimizer [53] is used in the proposed method for providing the adaptive learning rate with the learning scheduler. We selected the epoch, batch size, and optimizer for training the proposed models through empirical evaluation, demonstrating improved performance on the training dataset. The DNN system with the mentioned hyperparameters is trained on the system with the specifications shown in Table 3. The image input size and the training batch size are considered based on the hardware utilization in the DNN layers. The computation involved increases exponentially for a larger batch size and input image size, which may lead to system failure.

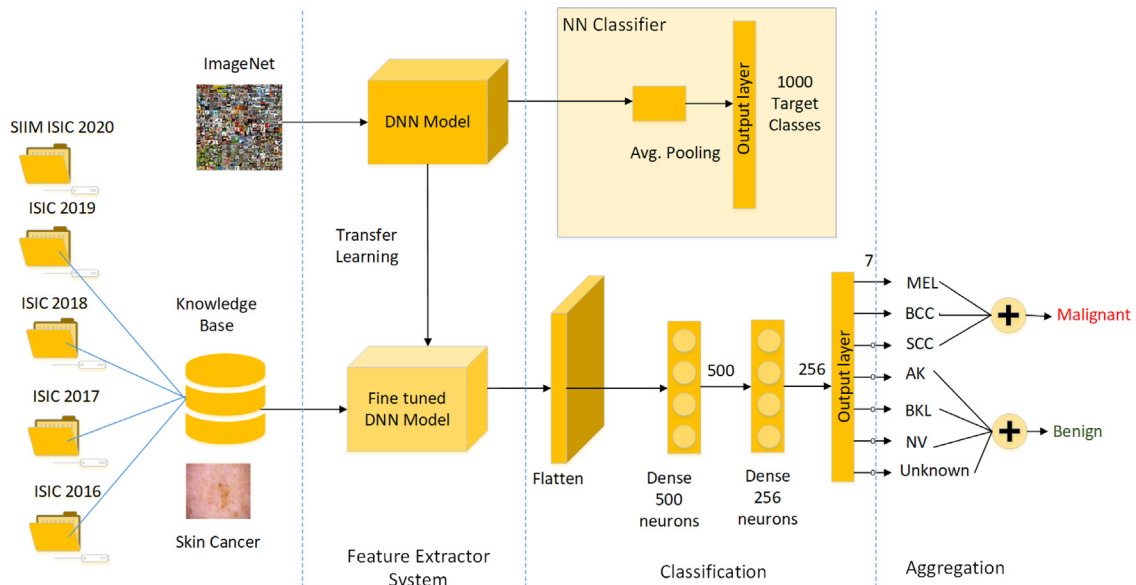


Fig. 2. The proposed DNN layered model for skin cancer detection.

Table 4

Performance metrics for evaluating the performance of models used for binary classification.

Performance measure	Mathematical representation	Range (Min, Max)
Accuracy (%)	$\frac{TP + TN}{TP + TN + FP + FN} * 100$	[0,100]
Sensitivity (TPR)	$\frac{TP}{TP + FN}$	[0,1]
Specificity (TNR)	$\frac{TN}{TN + FP}$	[0,1]
FPR	$\frac{FP}{TN + FP} = 1 - TNR$	[0,1]
Precision	$\frac{TP}{TP + FP}$	[0,1]
Recall	$\frac{TP}{TP + FN}$	[0,1]
F1 score	$2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$	[0,1]

Where TP : True positive, TN : True negative, FP : False positive, and FN : False negative.

2.3. Performance measures

The skin lesion classification models are evaluated using standard metrics, including accuracy, sensitivity, specificity, F1 score, precision, recall, and AUC score. The mathematical formulation of these measures is presented in Table 4. These metrics are computed by constructing confusion matrices based on the test dataset. The models are evaluated across the various datasets for performance evaluation.

In the proposed binary classification method used for skin cancer detection, the malignant class is considered the positive class, and the benign class is the negative class for a few reasons. Firstly, to provide priory to detect malignant skin lesions as they are more dangerous and have a higher potential for causing harm to the patient. The DNN system is trained to prioritize detecting and correctly classifying malignant images by designating the malignant class as positive. Secondly, it is more critical in medical applications to minimize false negatives rather than positives. False negatives, where a malignant lesion is classified as benign, can have severe consequences as they can delay diagnosis and treatment. Lastly, considering the dataset, the number of benign images is more than that of malignant images. In this scenario, if benign images are considered positive class, the model might be biased towards benign images, which can lead to a higher number of false negatives, which is not acceptable for a sound diagnosis system.

The model performance is evaluated using the performance metrics shown in Table 4. The accuracy is not the best measure for evaluating the model performance for binary classification [54]. The model's ability to distinguish between intra-classes is required for understanding the actual performance of the model. Thus, metrics like specificity, sensitivity, and AUC score are generally used as the gold standard for evaluating the models used for binary classification tasks. The specificity denotes the true positive rate (TPR), and it describes the ratio of the model's ability to predict the actual positive class based on the ground truth. At the same time, the sensitivity denotes the true negative rate (TNR) and describes the ratio of the model's ability to predict the actual negative class based on the ground truth. The probability inverse of sensitivity is denoted by the false positive rate (FPR), which describes the model's inability to predict positive class. The AUC score represents the area under the ROC curve plotted based on the relationship between TPR and FPR.

In multiclass classification, each targeted class is compared with the remaining class from the confusion matrix, and the individual TP, FP, FN, and TN can be calculated. The performance of the models is evaluated based on the performance metrics shown in Table 5 for the multiclass classification. In Table 5, the j represents the individual targeted class. The multiclass metrics are calculated as micro average, macro average, and weighted average. Micro average computes aggregated contribution to metrics value from all the classes. In comparison, the macro average computes independent metrics from the individual classes. Weighted average computes individual class contribution for the metric but is proportional to the weight or the number of samples in each class. The F1 score is calculated using the weighted average between the precision and recall metrics.

3. Results and discussions

The proposed model is fine-tuned and trained with transfer learning [39,50,51] with hyperparameter configurations shown in Table 2. The model architecture is implemented using the variant of EfficientNet, namely EfficientNet-B4 [33] and EfficientNetV2-M [47]. The model is trained with the knowledge base with seven labeled classes. The knowledge base is a redefined collection of dermoscopic images collected across the available dataset, and the information is preprocessed for enhancing the training method in DNN. The training is performed for 16 epochs, fine-tuning with Adam optimizer at a learning rate of 3×10^{-5} . The validation of the models is done after each epoch.

Table 5

Performance metrics for evaluating the performance of models used for multiclass classification.

Performance measure	Mathematical representation	Range (Min, Max)
Categorical accuracy (%)	$\frac{\sum_{c=1}^n TP_c}{N_{images}} * 100$	[0,100]
Micro avg. precision	$\frac{\sum_{c=1}^n TP_c}{\sum_{c=1}^n TP_c + FP_c}$	[0,1]
Macro avg. precision	$\frac{\sum_{c=1}^n \left[\frac{TP_c}{TP_c + FP_c} \right]}{N_{class}}$	[0,1]
Weighted avg. precision	$\frac{\sum_{c=1}^n N_c * \left[\frac{TP_c}{TP_c + FP_c} \right]}{\sum_{c=1}^n N_c}$	[0,1]
Micro avg. recall	$\frac{\sum_{c=1}^n TP_c}{\sum_{c=1}^n TP_c + FN_c}$	[0,1]
Macro avg. recall	$\frac{\sum_{c=1}^n \left[\frac{TP_c}{TP_c + FN_c} \right]}{N_{class}}$	[0,1]
Weighted avg. precision	$\frac{\sum_{c=1}^n N_c * \left[\frac{TP_c}{TP_c + FN_c} \right]}{\sum_{c=1}^n N_c}$	[0,1]

Where 'c' denotes the individual target classes in multiclass classification and 'N' denotes the total number.

The best model is the minimum validation loss during the training phase. The training performance of the model is tabulated in Table 6. The EfficientNetV2-M achieved the highest training accuracy of 94.10% and arrived at an improved minimum validation loss of 0.34 in the 11th epoch. In the case of EfficientNet-B4, the improved validation loss of 0.41 is attained at 16th epoch. The training accuracy of the best model using EfficientNet-B4 is 89.92%. Thus, the EfficientNetV2-M model has learned the features of the skin lesions at a faster rate with higher accuracy than EfficientNet-B4. Even though the EfficientNetV2-M and EfficientNet-B4 are both variants of the EfficientNet family, which is designed to be computationally efficient while maintaining high accuracy. From Table 6, it can be observed that the EfficientNetV2-M model took more training time than EfficientNet-B4. The EfficientNetV2-M [47] has more learnable parameters than EfficientNet-B4 [33]; thus, it requires slightly more training time. However, the specific computation requirements will depend on the size of the dataset and the specific hardware and software framework being used. Thus, it is important to note that EfficientNetV2-M is a larger network and may achieve better accuracy than EfficientNet-B4, but at the cost of longer training time and higher computational resources.

At the initial stage of model evaluation, the trained model performance is evaluated on multiclass datasets (ISIC 2020 [34], ISIC 2019 [26,28,29], and HAM10000 [26]). The data distribution for the target classes in the dataset is discussed in Section 2.1. The evaluation of multiclass classification helps to investigate the model nature to distinguish the intra-class information. The evaluation process provides insights into the DNN system's understanding of the hidden features describing individual skin lesions. The resulting performance metrics of the proposed model for the multiclass evaluation are tabulated in Table 7. The metrics used are micro, macro, and weighted averages. The F1 score is calculated from precision and recall. The multiclass performance evaluation shows that the EfficientNetV2-M model achieved the highest accuracy and F1 score over the entire collection of the test datasets. The DNN model using EfficientNetV2-M achieved categorical accuracy of 97.62%, 95.49%, and 94.8% with micro average F1 scores of 0.98, 0.95, and 0.95 on ISIC 2020 [34], ISIC 2019 [26,28,29], and HAM10000 [26] datasets, respectively for multiclass skin lesion classification. Micro average and weighted average F1 scores are highest when the model is evaluated with ISIC 2020 [34] test dataset. However, this evaluation has a low macro F1 score. A lower macro F1 score is because of the high imbalance in individual classes in ISIC 2020 [34] dataset.

The macro F1 score is the highest, with a value of 0.92 for the model evaluated with the ISIC 2019 [26,28,29] test dataset.

Table 8 compares the proposed modified EfficientNetV2-M with the results reported by recently developed state-of-the-art multiclass DL-based classification models using HAM10000 [26] and ISIC 2019 [26,28,29] test datasets. On the HAM10000 [26] dataset, the proposed modified EfficientNetV2-M achieved the highest accuracy of 94.80%, the highest among the compared DL models. Also, on the ISIC 2019 [26,28,29] test dataset, the proposed modified EfficientNetV2-M obtained a multiclass classification accuracy of 95.49%, which is the highest accuracy obtained.

The probability distribution of the multiclass is aggregated to binary-level classification in the proposed method. The binary level classification groups the skin lesions broadly into malignant and benign. This grouping helps in detecting skin cancer. The model performance on binary-level classification across the test datasets is tabulated in Table 9. The F1 score in binary classification is higher for the proposed model in comparison to the F1 score in multiclass classification. Thus, aggregating individual classes helps the model understand the skin lesion better and provides better performance. The results show that the EfficientNetV2-M has the highest performance metrics in all the test datasets, similar to the multiclass classification. The DNN model using EfficientNetV2-M achieves the highest accuracy of 99.23%, 97.06%, and 95.95% with AUC scores of 0.98, 0.99, and 0.98 in ISIC 2020 [34], ISIC 2019 [26,28,29], and HAM10000 [26] test datasets, respectively. The higher AUC score represents the model's ability to distinguish the actual positive class.

The ROC curves are plotted to compare the modified EfficientNet-B4 and EfficientNetV2-M model's performance across various test datasets. The modified EfficientNetV2-M outperforms the modified EfficientNet-B4 model on all three datasets. The AUC score is highest with a value of 0.99 when the EfficientNetV2-M model is evaluated with ISIC 2019 [26,28,29] test dataset. The AUC score is lowest with a value of 0.96 for the EfficientNet-B4 model when evaluated on ISIC 2020 [34] test dataset. The modified EfficientNetV2-M model obtained the highest macro F1 score of 0.92 in multiclass classification and achieved the highest AUC score of 0.99 in binary classification on the ISIC 2019 [26,28,29] test dataset (see Fig. 3).

The Gradient-weighted Class Activation Mapping (Grad-CAM) [62] is a technique that can be used to generate heatmaps that highlight the regions in an image that are most important for a particular classification decision made by the DNN models [63]. The important neuron weight (a_f^c), as shown in Eq. (1), is used to calculate the heatmap. It is obtained by the global average pooling of the gradients calculated during the backpropagation process. The gradients are computed based on the output of the final convolutional layer of the DNN, and they indicate how the output of the DNN changes for the input image. The global average pooling is performed over all the gradients, which results in a single importance weight value for each neuron in the final convolutional layer.

$$a_f^c = \frac{1}{P} * \sum_m \sum_n \frac{\partial y^c}{\partial A_{mn}^f} \quad (1)$$

In Eq. (1), ' y^c ' represents the output of the image classification network, ' c ' represents the score for individual target class (malignant or benign), ' P ' is the total number of pixels in the feature map, and ' f ' represents the number of channels in the convolutional layers or the feature maps to compute the Grad-CAM map.

The gradients of the last convolution layer in the feature extraction layer and is denoted by $\frac{\partial y^c}{\partial A_{mn}^f}$, where ' m ',' n ' are the spatial information of pixels and ' A_{mn}^f ' is the obtained feature maps from the layer. Based on the neuron importance weight, the class discrimination for Grad-CAM localization map ' $M_{GradCAM}^c$ ' is computed as the forward activation of ReLU for weighted feature maps ' A^f ' as shown in Eq. (2).

$$M_{GradCAM}^c = ReLU \left(\sum_f a_f^c \cdot A^f \right) \quad (2)$$

Table 6

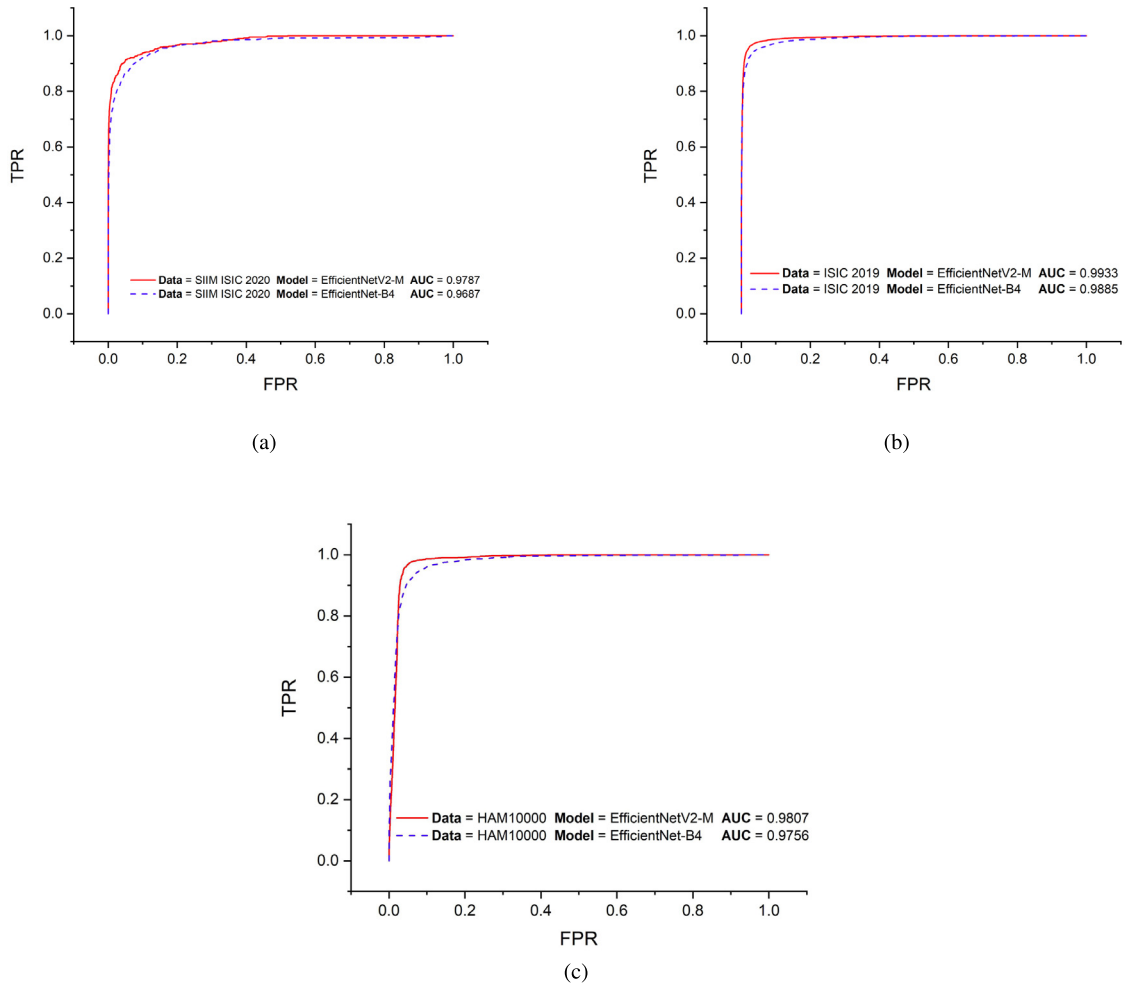
Training performance of the models with fine-tuning and transfer learning on the knowledge base.

DNN model	Parameters (Million)	Training			Validation	
		Time (h)	Acc. (%)	Loss	Acc. (%)	Loss
EfficientNet-B4, 2020 [33]	19	30	89.92	0.30	88.19	0.41
EfficientNetV2-M, 2021 [47]	55	36	94.10	0.18	90.19	0.34

Table 7

Performance evaluation of the proposed models in multiclass classification for skin lesions.

DNN model	Dataset	Acc. (%)	Precision			Recall			F1 score		
			Micro	Macro	Weighted	Micro	Macro	Weighted	Micro	Macro	Weighted
EfficientNet-B4	ISIC 2020	96.95	0.97	0.51	0.97	0.97	0.42	0.97	0.97	0.46	0.97
	ISIC 2019	93.17	0.93	0.89	0.93	0.93	0.89	0.93	0.93	0.89	0.93
	HAM10000	92.28	0.92	0.77	0.94	0.92	0.69	0.92	0.94	0.70	0.93
EfficientNetV2-M	ISIC 2020	97.62	0.98	0.53	0.98	0.98	0.47	0.98	0.98	0.49	0.98
	ISIC 2019	95.49	0.95	0.93	0.96	0.95	0.92	0.95	0.95	0.92	0.95
	HAM10000	94.80	0.95	0.82	0.97	0.95	0.71	0.95	0.95	0.74	0.95

**Fig. 3.** ROC plot for classification of skin cancer (malignant and benign) by EfficientNet-B4 and EfficientNetV2-M for various datasets: (a) ISIC 2020 dataset, (b) ISIC 2019 dataset, and (c) HAM10000 dataset.

The ReLU activation function is applied to the feature maps to ensure that only the positive contributions to the class of interest are considered, resulting in a heatmap for the class. The Grad-CAM map is then upsampled to the size of the input image to show which regions of the image were most important for the prediction.

The Grad-CAM localization for the proposed DNN system using modified EfficientNetV2-M is evaluated on dermoscopic images shown in Fig. 4. The visualization of the Grad-CAM map shows that the proposed DNN can effectively discriminate malignant and benign features.

It helps to provide an explainability to the proposed DNN model and insights into the features learned by the modified EfficientNetV2-M model.

4. Conclusion

This paper presents DNN models based on modified EfficientNet-B4 and EfficientNetV2-M models for classifying malignant and benign skin lesions from dermoscopic images. A knowledge base of 58,032

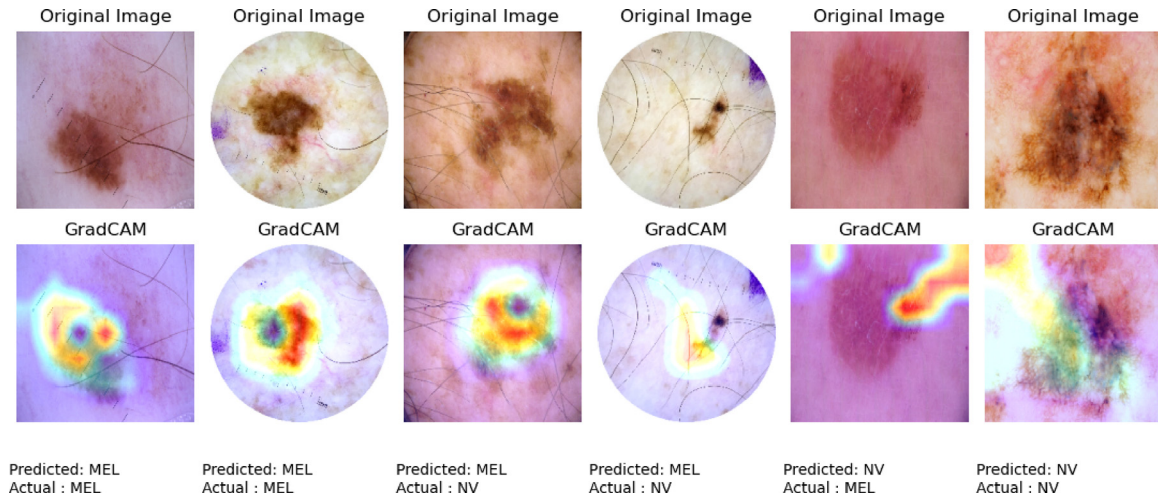


Fig. 4. GradCAM localization of region from the DNN model based on EfficientNetV2-M for skin cancer detection.

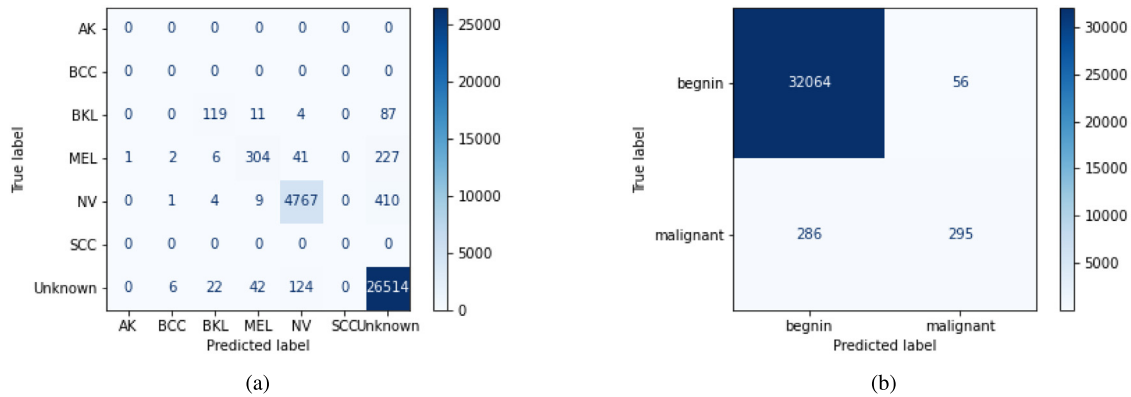


Fig. 5. EfficientNet-B4 model performance on ISIC 2020 dataset: (a) multiclass classification (AK, BCC, BKL, MEL, NV, SCC, and Unknown), and (b) binary class classification (malignant and benign).

Table 8

Comparison of the proposed modified EfficientNetV2-M against recent state-of-the-art deep learning-based multiclass classification models.

Reference	Year	Method	Datasets	Accuracy (%)
Tahir et al. [55]	2023	DSCC-Net	HAM1000	94.17
Sukanya et al. [41]	2023	Deep Belief Net		94.00
Chaturvedi et al. [30]	2023	ResNetXt-101		93.20
Zafar et al. [42]	2023	DeepLabV3+		92.07
Alam et al. [56]	2022	RegNetY-320		91.00
Ali et al. [38]	2022	EfficientNet-B4		87.90
Khan et al. [57]	2021	Mask RCNN		88.50
Proposed Modified EfficientNetV2-M				94.80
Zafar et al. [42]	2023	DeepLabV3+	ISIC 2019	91.70
Gilani et al. [46]	2023	Spiking VGG-13		89.50
Zhou et al. [58]	2023	FixMatch-LS		91.81
Naeem et al. [59]	2022	SCDNet		92.91
Pacheco et al. [60]	2020	VGG-16		82.50
Kassem et al. [61]	2020	GoogLeNet		94.92
Pacheco et al. [31]	2019	Ensemble CNN		90.10
Proposed Modified EfficientNetV2-M				95.49

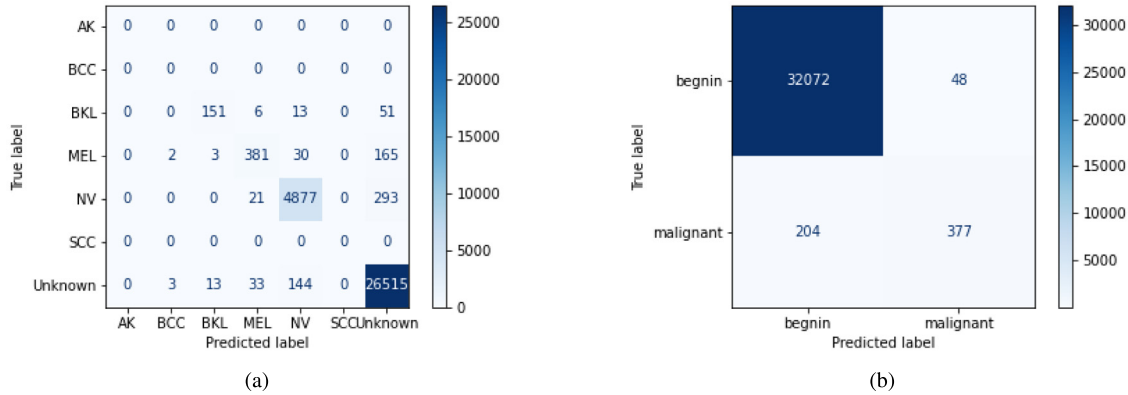
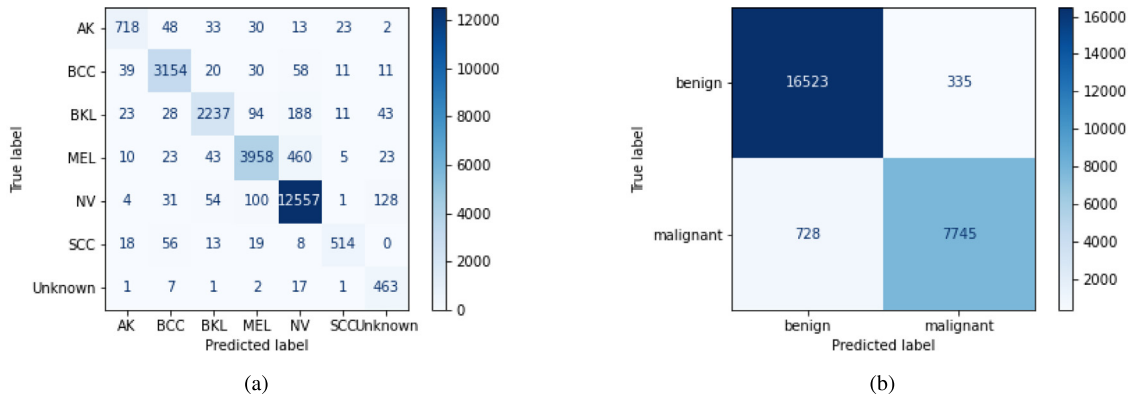
refined dermoscopic images from the ISIC 2020, ISIC 2019, and ISIC 2018 databases were used to train and evaluate DNN models for skin lesion classification. Transfer learning and data augmentation techniques were applied to enhance the generalizability and prevent overfitting of the models. The models were evaluated for multiclass and binary classification tasks. The results showed that the EfficientNetV2-M architecture outperformed other state-of-the-art DL-based classification models in terms of accuracy and AUC score.

The current study has used fewer images of malignant lesions to train the model, which may result in the model being biased toward classifying them as benign. It would be interesting to evaluate the performance of DNN models trained using images generated by Generative Adversarial Networks (GANs). Furthermore, incorporating ensemble learning and attention-based modeling can enhance the performance of the DNN models. In the future, an ensemble of attention-based models could be trained, each focusing on different regions of the input image. Their predictions could be combined to make the final diagnosis. This approach could improve the robustness and generalization of the model, reducing the risk of misdiagnosis. Also, the DNN models can be trained with a tissue-level knowledge base to incorporate into diagnosis devices used in computational pathology. The proposed method can potentially be used as assistive technology in computer-aided diagnosis tools for dermatologists.

Table 9

Performance evaluation of the proposed models in binary classification for skin cancer detection.

DNN model	Dataset	Acc. (%)	Performance metrics on binary classification (malignant and benign)					
			Sensitivity	Specificity	Precision	Recall	F1 score	AUC score
EfficientNet-B4	ISIC 2020	98.95	0.84	0.99	0.84	0.51	0.63	0.9687
	ISIC 2019	95.8	0.96	0.96	0.96	0.91	0.93	0.9885
	HAM10000	94.86	0.83	0.97	0.83	0.86	0.84	0.9756
EfficientNetV2-M	ISIC 2020	99.23	0.89	0.99	0.89	0.65	0.75	0.9787
	ISIC 2019	97.06	0.96	0.97	0.96	0.95	0.95	0.9933
	HAM10000	95.95	0.83	0.99	0.83	0.94	0.88	0.9807

**Fig. 6.** EfficientNetV2-M model performance on ISIC 2020 dataset: (a) multiclass classification (AK, BCC, BKL, MEL, NV, SCC, and Unknown), and (b) binary class classification (malignant and benign).**Fig. 7.** EfficientNet-B4 model performance on ISIC 2019 dataset: (a) multiclass classification (AK, BCC, BKL, MEL, NV, SCC, and Unknown), and (b) binary class classification (malignant and benign).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

Acknowledgments

The authors thank the National Institute of Technology Puducherry, India, for providing computational resources. We are thankful to Dr. M. Vipin Das, Dermatologist, Kerala Health Services, Trivandrum, Kerala, India, for giving his valuable suggestions and for validating the results obtained from the proposed models.

Appendix A. Confusion matrices for skin lesion classification (multiclass & binary class classification) by the proposed model in ISIC 2020 dataset.

See Figs. 5 and 6.

Appendix B. Confusion matrices for skin lesion classification (multiclass & binary class classification) by the proposed model in ISIC 2019 dataset.

See Figs. 7 and 8.

Appendix C. Confusion matrices for skin lesion classification (multiclass & binary class classification) by the proposed model in HAM10000 dataset.

See Figs. 9 and 10.

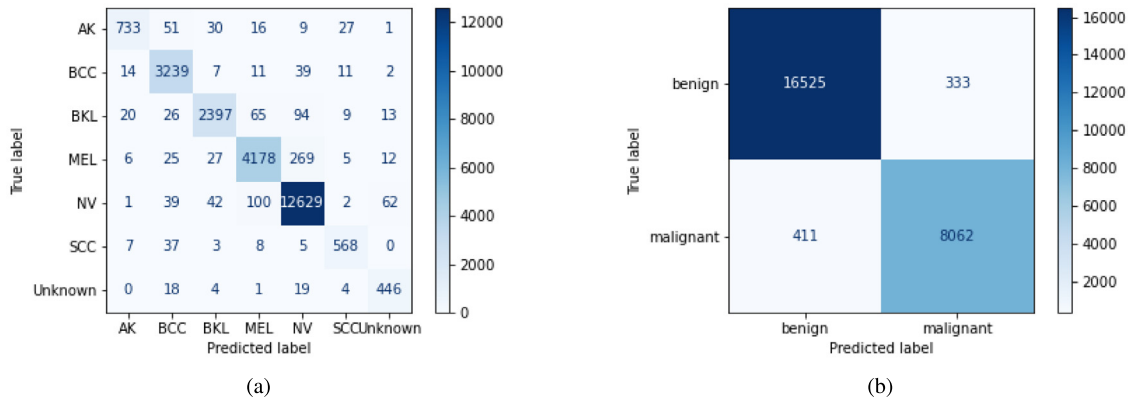


Fig. 8. EfficientNetV2-M model performance on ISIC 2019 dataset: (a) multiclass classification (AK, BCC, BKL, MEL, NV, SCC, and Unknown), and (b) binary class classification (malignant and benign).

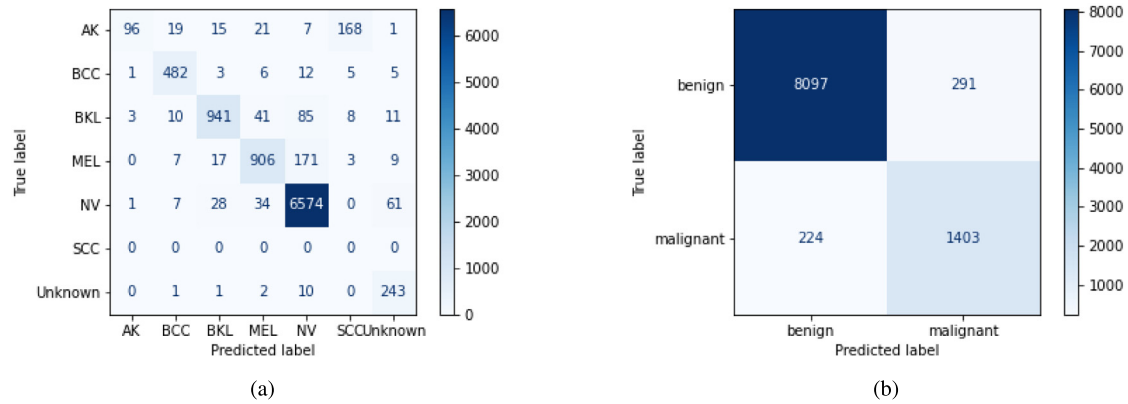


Fig. 9. EfficientNet-B4 model performance on HAM10000 dataset: (a) multiclass classification (AK, BCC, BKL, MEL, NV, SCC, and Unknown), and (b) binary class classification (malignant and benign).

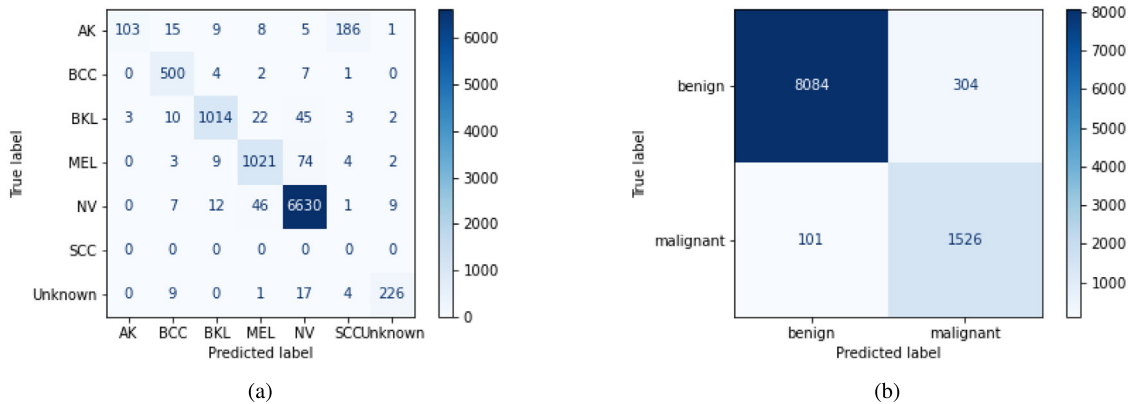


Fig. 10. EfficientNetV2-M model performance on HAM10000 dataset: (a) multiclass classification (AK, BCC, BKL, MEL, NV, SCC, and Unknown), and (b) binary class classification (malignant and benign).

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